

BENJAMIN LAM

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Work Experience

Tesla, Inc.

Jan 2025 – Present

Drive Unit Test Engineer

- **Established processes for drive inverter fault-response testing**, authoring the first test specifications and procedures and enabling systematic expansion of this test category. Work cross functionally with firmware, safety, motor controls, motor EM simulations, and electrical board design teams
- Characterized inverter electrical behavior during fault events, evaluating gate drive signals, phase currents, and protection timing to inform drive unit replacement decisions and mitigate damage. Automated test execution and data acquisition (Oscilloscope automation, Python), achieving ~10x faster execution and improved data traceability
- Evaluated edge-case electrical readiness of inverter design changes, assessing worst-case current and protection margins. Identified reliability risks in cost-down modifications and drove mitigation prioritization, validation testing, and risk assessment
- **Developing a real-time fault detection system** for all lab test stations to preserve critical teardown data and reduce repeat testing after catastrophic failure. **Employed machine learning methods (Python)** to account for large variability across test profiles and motor types. **Integrating into existing Python/NI Veristand test stand codebase**
- Managing testing operations for a major drive unit program, responsible for test planning, coordination with design teams, execution, test setup assembly, and analysis to support change validation and failure analysis
- Owned root cause analysis of a critical component failure, creating new electrical and thermal metrics to evaluate hardware and firmware solutions and recommending a firmware fix deployed fleetwide, improving system safety and reliability

Ceder Group (UC Berkeley)

Aug 2022 – Dec 2024

Graduate Student Researcher (Battery Materials)

- Designed + built **automated in-situ XRD sample loader robot** (Flask/Python/I²C) enabling high-throughput characterization for integration into our lab's autonomous synthesis lab [\[Github\]](#)
- Designed + built **custom Li-air battery test system** controlled with Flask/Python/I²C to cycle cells in PID-controlled relative humidity/gas composition. Designed Swagelok-type cell for airtight sealing, assembly/control of electronic sensors and actuators, and all software. **Added Li-air battery test capabilities to our lab** (Python, C++, JavaScript, HTML) [\[Github\]](#)
- Identified the root-cause failure mechanism of Li-NASICON solid-state conductors in humid air battery cells, **eliminating all phosphate chemistries from consideration due to inherent instability in alkaline conditions**. Conducted degradation research using time-resolved XRD, SEM/EDS, and ICP-OES, with data analysis in MATLAB. Experimental results supported by DFT surface simulations and ab-initio Pourbaix diagrams. [\[Advanced Energy Materials \(2024\), First Author\]](#)

Projects

- *Drone Project (Ongoing)*: Writing flight control firmware for the nRF52840 (**Nordic Connect SDK/Zephyr RTOS in C**), implementing real-time sensor fusion (I²C accel/gyro) and Bluetooth wireless control
- *Microfluidics PCB*: Developed a fully integrated solenoid array controller for microfluidic chip testing, with 32 parallel solenoid drivers, real-time pressure sensing, on-board power conversion, and CAN/I²C communication.

Education

University of California, Berkeley

Dec 2024

MS in Materials Science Engineering, **GPA: 3.97**

- *MPC Final Project: Automated path generation and in-route correction for parking* a 2-wheeled rover. Simulations with wheel slip considered to evaluate control robustness to external disturbances

University of California, Irvine

Jun 2022

BS in Mechanical Engineering and Materials Science Engineering (Double Major), **GPA: 3.98**

- Dean's Honor List (All quarters), Graduated Summa Cum Laude
- *UCI Solar Car Battery Lead*: Directed recruitment and selection of new electrical team members
- *Bowman Lab Research*: Custom MATLAB library developed for processing of EIS datasets, contributed to **2 papers**: [\[ACS Applied Nano Materials \(2023\)\]](#) [\[Advanced Functional Materials \(2024\)\]](#)
- *Hui Lab Research*: Developed a simulated model for a microfluidic valve. **Made new valve design** with truncated valve floor helps improve size variation of generated droplets (down to $\sigma = 1\%$, **3x improvement**) [\[Micromachines \(2022\)\]](#)

Skills

Electrical: Oscilloscope, Power Analyzer, PCB Prototyping, High voltage experience, Inverters, AC motor controls, Fault response

Programming: Python (3+ years), MATLAB (4+ years), C, Java (2 years)

Mechanical: SolidWorks (CSWA Certified), Autodesk Fusion 360, Autodesk CFD, Dynamometer lab

Materials Science: XRD, SEM/EDS, ICP-OES/MS, EIS, Aqueous Chemistry, TGA/DSC, Chemical Lab Experience (3+ years)